Drug Resistance Mutations Affect on HLA Binding in HIV-1 Epitopes

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1 Introduction & Background

Human Immunodeficiency Virus Type 1 (HIV-1), a virus transmitted by body fluids, capable of shutting down the host immune system if not treated immediately [23]. As of 2020, an estimated 37 million people around the globe suffer from HIV-1 [28]. Furthermore, HIV-1 continues to spread, where it is estimated an additional 1.7 million global citizens acquire HIV-1 on average each year [28]. Despite having been discovered over 4 decades ago in 1981 [25], HIV-1, the cause of Acquired Immune Deficiency Syndrome (AIDS), remains an incurable disease and evades any form of vaccination or cure [28] where the best treatment option for patients of HIV-1 is to undergo antiretroviral therapy (ART) treatment [17]. With ART, HIV-1 associated mortality rates have been successfully reduced ultimately saving millions, and changing AIDS from a deadly disease, to one where the patients may typically live a full life [28]. However, it is critical to accept ART not as a complete solution, rather, as a temporary solution. ART itself remains an imperfect treatment with problems ranging from rebound due to patients not fulfilling the required periodic treatment plan [2], to constrained availability of treatment [22], to problematic cost of treatment [10, 25].

1.1 Drug Resistance

Amongst these problems, however, there underlies a biologically critical change HIV-1 is undergoing common to many pathogens where a treatment is not potent enough to cure is the gold standard and thus deterring the progress of ART. Specifically, in recent years, with advent of ART, HIV-1, has been slowly but surely developing resistance to many modern ART treatment related drugs [3, 4, 7].

To elaborate on HIV-1 drug resistant mutations (DRMs), it is typical that a single amino acid variation coded for in the HIV-1 genome is not enough to induce resistance towards antiretroviral drugs (ARDs) [3]. Furthermore, within a given drug classification, it is rather common to see cross-resistance where the increased resistance towards one particular drug increases the resistance towards another [3]. Despite such cross-resistance prevalence within the same drug class, the same cannot be said across different drug classes where current research seems to indicate that it is more difficult [11] for such cross-resistances, however, not impossible [3, 13, 24]. This lower prevalence of cross-class drug resistance is a key reason for the use of multiple ARDs of differing classes during modern ART treatment plans [6]. Notably, such DRMs seem to typically result in a fitness penalty to the host strain of the DRMs [3].

1.2 Immune Escape

HIV-1 is fascinating in that it is one of the few viruses in modern day that both is highly prevalent in the population, but also eludes cures. Specifically, Cytotoxic T Cell (CTL), also known as CD8+ T cells, are well known to be a host's main antagonist to theHIV-1 virus [5]. Then, it goes to reason, that HIV-1, with a

relatively unstable genome capable of high rates of proliferation [21], since it's initial entry into the human population, has developed various mechanisms of circumvent, or escape, an effective CTL response. One such mechanism is via certain specific immune escape mutations (IEMs) that result in the suppressing of displaying specifically HIV-1 associated antigens thus avoiding apoptosis via natural killer T cells [27]. Similar DRMs, IEMs such IEMs also seem to introduce a fitness penalty to the HIV-1 strain [12, 31].

1.3 Motivation for Study

Both mutations, independent of each other are relatively thoroughly studied. However, at the time of writing, there exists a minimal amount of studies looking into the intersection between both DRMs and IEMs. In other words, since both forms of mutations incur fitness penalties, an area of interest may be how DRMs affect the mutation based immune escape process via epitopic modifications. The few studies emphasizing such a topic discusses various DRMs that both corroborate the increase of immunogenicity of HIV-1 [1, 15], and a decrease in immunogenicity, depending on varying factors [16, 18]. The results of these studies are reasonable as DRMs occurring within epitope regions, i.e., regions that affect the presentation via human leukocyte antigen class I (HLA-I) may modulate the fitness penalty due to directly affecting a immune escape mechanism. This study hopes to extend the knowledge base surrounding such interplay between DRMs and IEMs as such knowledge may serve foundational to improving treatment regiment by selecting drugs that produce strong fitness penalties and the future development of HIV-1 vaccines, as explained by previous studies [16, 18].

2 Materials & Methods

2.1 Materials

Analysis of the interplay between DRMs, and IEMs were performed upon datasets upon the Los Alamos HIV databases (LANL-HIV) [14] and the Stanford HIV Drug Resistance Mutation Database (DRM-DB) [20]. NetMHCPan version 4.1 and an in-house python script were used in combination for accommodating large data throughput [19]. Within the Python script, SciPy and Pandas data processing libraries were used for the analysis [26, 29]. Figures were generated with use of Matplotlib and Seaborn [8, 30].

The LANL-HIV dataset of known HIV-1 CTL epitopes including the peptide sequence and position relative to the gene. Datasets were individually downloaded by querying for integrase, p24, protease, and Reverse Transcriptase (RT) while all other search parameters were left as the most broad casting option i.e., selecting the "ALL" option when present. The DRM-DB amongst other tools, and other datasets, contains a scoring mechanism for known drug resistance inducing mutations in the HIV-1 genome [20]. Spreadsheets of such scoring for the different drugs with respect to a given single mutation were downloaded. NetMHCPan version 4.1 was used to predict binding against different HLA-I alleles [19]. Data ingress and egress, manipulation and vectorized operations were performed by Pandas and the SciPy library was used for statistical associations [26, 29]. Lastly, Seaborn was used for the automated generation and regression figures by making use of Matplotlib [8, 30].

2.2 Methods

All aforementioned datasets were imported from respective comma separated values (CSV) sheets downloaded from their respective databases. This was then followed by the generation of new epitopes based off the set of known epitopes, and the set of drug resistant mutations. All epitopes, natural or generated were then run through NetMHCPan 4.1 to predict their binding level towards various HLA-I alleles. Such binding levels were fit into categories. Hypothetical epitopes were generated from the naturally occurring epitopes (NEPIs). All generated epitopes were cleaned and filtered to increase the number of interpretable natural epitopes and reduce reduce computation time respectively before running the dataset in NetMHCPan [19]. DRMs and their associated resistance scores were summed in accordance to the guidelines of the DRM-DB. The resistance scores and DRM scores were then categorized according to their respective guidelines from DRM-DB. Association between the presence of DRMs and the binding level with various HLA-I alleles were tested and p-values were aggregated.

The first step involved downloading the data and parsing the data into Python. Data was procured from both LANL-HIV and DRM-DB as described in the materials section. DRM were further parsed and merged into a large, master spreadsheet where the drugs irrelevant to a mutation were set to have

a DRM of 0. Similarly, known N-EPIs were also parsed. Unused columns of metadata were dropped and epitopes with the exact identical sequences were dropped immediately as to ensure that any associations found would be purely based on their occurrence in a epitope, and not biased by the frequency in a population. Furthermore, the LANL-HIV contained synthetic epitopes created by concatenating two epitopes together. Such epitopes did not have well-defined bounds and thus, also did not have well-defined DRMs [14]. Such junction-based epitopes had their junctions removed and the original epitope restored by selecting the portion of the junction epitope based on the annotated bound.

The unique set of naturally occurring HIV epitopes then had variants generated naively simply by generating all possible different combinations of mutations that could be added to a naturally occurring epitope was created. These epitopes were then added to a new flat database with their original DRMs, i.e., mutations that existed before any modification, and the added mutations. Notably, due to some epitopes having many possible combination of DRMs, constraints were added such that generated epitopes that resulted in a DRM score lower than 15 was removed. This corresponds to DRM-DBs categorization of having none, or potentially a very low level drug resistance (Table 2). Additionally, some N-EPI had many peptide positions with known DRMs and thus, a ranked threshold was applied where the DRM scores were used to rank the generated epitopes and such epitopes that ranked higher than 1000 were removed. I.e., the top 1000 highest DRM scoring generated epitopes were kept.

Then, the produced aggregate of both N-EPIs and the generated epitopes had binding levels with all HLA-I alleles that were recognized by NetMHCPan 4.1 (table 1) predicted and cataloged [19]. The resulting database had DRM scores and eluted ligand percentile rank (%EL-Rank) [9] calculated and mapped according to table 2 and to table 3 respectively (for more information on this metric, see original software paper for version 4.0) [9, 19]. Furthermore, changes in binding level were calculated by comparing the N-EPIs with the generated ones.

Lastly, all combinations between all sets of DRMs, and the empty set of DRMs, i.e., the lack of DRMs, were tested for association with all different levels of binding, that is to say, these combinations of DRMs were tested against the occurrences of weak, strong, and no binding level epitopes. The Fisher's exact test was used due to the relatively small sample count of epitopes with the same set of DRMs with unique epitopes. This process generated a list of different p-values for each possible combination. This list of p-values was then filtered by a $\alpha = 0.05$.

Resulting tables were generated via Pandas and resulting figures were generated with Seaborn and Matplotlib [8, 26, 30]. Further post-analysis filtering and selection of datasets were also performed by these libraries.

Supported HLA Variants
HLA-A02:01
HLA-A01:01
HLA-A03:01
HLA-A24:02
HLA-A26:01
HLA-B07:02
HLA-B08:01
HLA-B27:05
HLA-B39:01
HLA-B40:01
HLA-B58:01
HLA-B15:01

Table 1: The supported list of HLA alleles for which NetMHCPan is capable of predicting %EL-Rank.

Resistance Category	DRM Score (Inclusive)
No drug resistance	0 to 9
Potential low-level drug resistance	10 to 14
Low-level drug resistance	15 to 29
Intermediate resistance	30 to 59
High-level resistance	60 and above

Table 2: The categorical levels of drug resistance based on the DRM score assigned for an entire sequence as defined by DRM-DB.

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Resistance Category	DRM Score (Inclusive)
Strong Binding	Less than 0.5
Weak Binding	Less than 2, greater than or equal to 0.5
No Binding	Other

Table 3: The categorical levels of HLA binding based on the percentile rank predicted by NetMHCPan 4.1 where both the metric and the thresholds were advised by the original paper and authors of the software [19].

3 Results

Overall, a total of 390,816 unique epitopes were used for the following interpretation and analysis. From these 390,816 epitopes, 202,740 unique sets of epitopic DRMs were generated.

3.1 Comparing Distribution of Binding Strength

Figure 1 is a histogram where the vertical axis is the proportion of all DRM generated epitopes binned based on %EL-Rank with 50 bins resulting in every 2 percentiles being encapsulated in one bin. The N-EPI and their %EL-Rank are overlaid in orange with identical binning parameters. See that by table 3, the first bin contains all binding values. Such a figure indicates over all DRMs generated unique epitopes, there is shift of portion of epitopes towards being non-binding towards HLA-I. Furthermore, see the substantial drop in binding epitopes in contrast with the N-EPI.

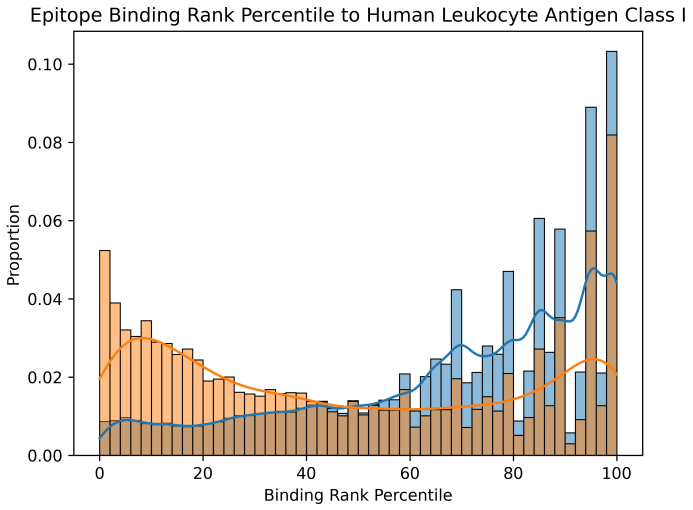


Figure 1: Histogram of generated epitopes based on drug resistant mutations binned by eluted ligand percentile rank.

However, filtering out results to include only generated epitopes with a maximum of 2 DRMs, the histogram becomes figure 2. Figure 2, in contrast with figure 1, produces a histogram where the distribution of DRM generated epitopes is much more similar to the distribution of the N-EPI.

Note in both figures 1 and 2, that there is proportionally more N-EPIs binding to HLA-I in contrast with the DRM generated set of epitopes.

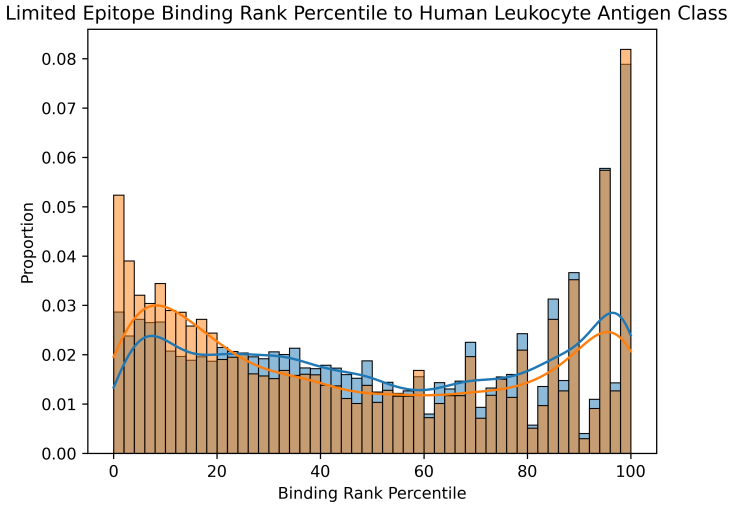


Figure 2: Histogram of generated epitopes with at most 2 drug resistant mutations with 50 bins by eluted ligand percentile rank.

3.2 Binding of Original Epitopes vs. Generated Epitopes

Despite the lack of any obvious relation between DRMs, when the data is stratified by drug class, and then stratified further by the HLA-I allele, we can see that some HLA-I variants for some drug classes do show a non 1:1 association between the DRM generated epitope, and the corresponding N-EPI. This can be seen in figure 3 where some HLA-I alleles do not grow in a 1:1 fashion. Namely, HLA-B07:02 (blue), and HLA-A24:02 both seem to have a few points of deviation from the standard 1:1 relation seen in the others. Furthermore, see that HLA-B07:02, amongst others, overall has many more epitopes within weak binding percentiles in contrast with many others.

3.3 Mutation Sets Associated with change in Binding

With regards to statistics and mutation set association results, we will look at the top 13 most significant results that contain dramatic binding changes, i.e., binding changes that go from no binding, to strong binding, or vice versa. Table 4 indicates the counts of sets of mutations occurring in relation to different binding strengths. Of note, see that of the top 13 most significant associations with dramatic changes in binding level, 11 results indicate a form of binding with HLA-I variant HLA-I-B07:02. Furthermore, of these results, all indicate that the insertion of such mutations result in an increase in binding where previously, there may have not been any form of binding, these mutations introduce strong binding with HLA-I.

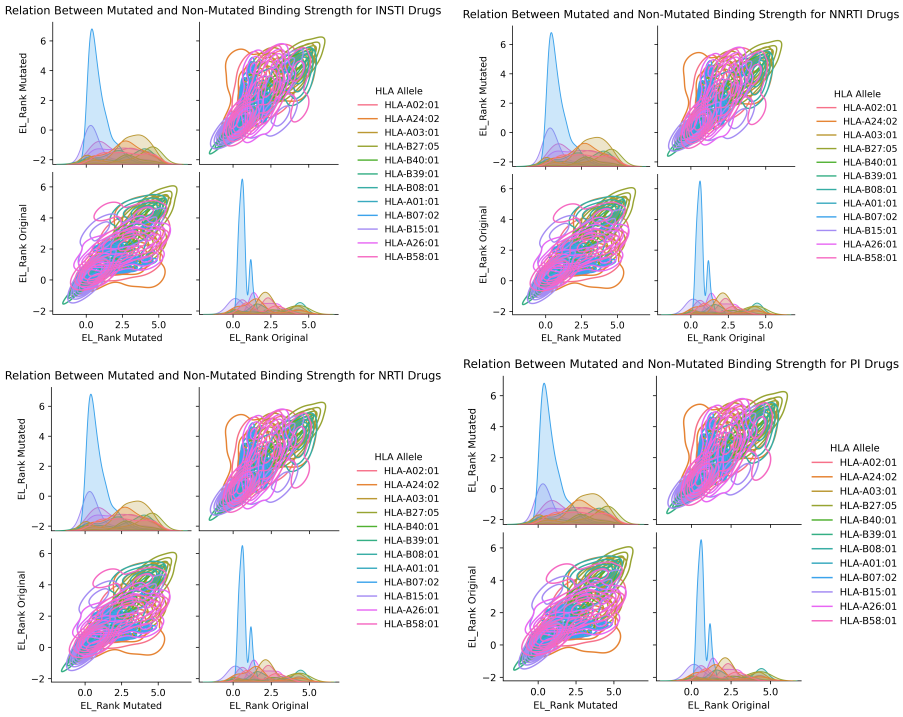


Figure 3: kernel density estimate plots of the relation between the change in binding affinity towards various human leukocyte antigen class I alleles. Only epitopes with at most 2 drug resistant mutations, had a original binding rank of less than or equal to 5, and has a mutated binding rank of less than or equal to 5 were included in the plotting.

Furthermore, see that for each result mutation set (table 5), the score of these sets cumulatively, have at least one drug of the respective class that is considered strongly drug resistant.

HLA Allele	HLA-B07:02	**2-Way Table**	**SB**	**NB**
Drug Target	Integrase	α **&** β*****	2	0
P-Value	0.00026	**No DRMs**	2	209
HLA Allele	HLA-B07:02	**2-Way Table**	**SB**	**NB**
Drug Target	Integrase	α **&** β†	2	1
P-Value	0.00078	**No DRMs**	2	209
HLA Allele	HLA-B15:01	**2-Way Table**	**SB**	**NB**
Drug Target	Integrase	**E92Q & Q95K**	3	8
P-Value	0.00080	**No DRMs**	2	215
HLA Allele	HLA-B15:01	**2-Way Table**	**SB**	**NB**
Drug Target	Integrase	**E92Q**	3	9
P-Value	0.00104	**No DRMs**	2	215
HLA Allele	HLA-B07:02	**2-Way Table**	**SB**	**NB**
Drug Target	Protease	**N88T & V82M**	2	0
P-Value	0.00111	**No DRMs**	1	71

*****: α and β may be of any of the following combinations (6): (Q**146**P, Q**148**R) (Q148H, Y143K), (Q148H, Y143R), (Q148K, Y143R), (Q148N, Y143R), (Q148R, Y143R)

†: α and β may be of any of the following combinations (4): (Q146P, Y143H), (Q146P, Y143K), (Q146P, Y143R), (S**147**G, Y143R)

Table 4: The top 13 most statistically significant associations between a given set of mutations existing in all unique variations of epitopes, with a change of eluted ligand percentile rank such that the categorical binding strength changes from strong binding (SB) to no binding (NB), or vice versa. Note that due to some of the associations having the same 2-way table, with the same properties other than the mutations, they were condensed to save space.

Drug Target: Integrase

HLA	Mutation Set	Binding	BIC	CAB	DTG	EVG	RAL
B07:02	Q146P, Q148R	SB, NB	25	70	25	120	60
B07:02	Q148H, Y143K	SB, NB	30	40	30	70	120
B07:02	Q148H, Y143R	SB, NB	30	40	30	70	120
B07:02	Q148K, Y143R	SB, NB	35	60	35	70	120
B07:02	Q148N, Y143R	SB, NB	5	25	5	25	75
B07:02	Q148R, Y143R	SB, NB	30	50	30	70	120
B07:02	Q146P, Y143H	SB, NB	5	40	5	70	60
B07:02	Q146P, Y143K	SB, NB	5	40	5	70	60
B07:02	Q146P, Y143R	SB, NB	5	40	5	70	60
B07:02	Q146G, Y143R	SB, NB	15	20	15	70	70
B15:01	E92Q, Q95K	SB, NB	15	20	15	70	70
B15:01	E92Q	SB, NB	10	15	10	60	30

Drug Target: Protease

HLA	Mutation Set	Binding	ATV/r	DRV/r	LPV/r	FPV/r
B07:02	Q146P, Q148R	SB, NB	25	0	25	15

HLA	Mutation Set	Binding	IDV/r	NFV	SQV/r	TPV/r
B07:02	Q146P, Q148R	SB, NB	30	60	15	10

Table 5: Drug resistance scores according to the Stanford HIV Drug Resistance Mutation Database corresponding to the top 13 most significant results. Both mutation set column and binding columns correspond to their respective information in table 4. Where SB indicates strong binding, and NB indicates a lack of binding. Note that the Protease table has been broken into two lines.

4 Discussion

Presently,

5 Acronyms

%EL-Rank eluted ligand percentile rank

AIDS Acquired Immune Deficiency Syndrome

ARD antiretroviral drug

ART antiretroviral therapy

CSV comma separated values

CTL Cytotoxic T Cell

DRM drug resistant mutation

DRM-DB Stanford HIV Drug Resistance Mutation Database

HIV-1 Human Immunodeficiency Virus Type 1

HLA-I human leukocyte antigen class I

IEM immune escape mutation

LANL-HIV Los Alamos HIV databases

N-EPI naturally occurring epitope

RT Reverse Transcriptase

References

[1] Juan Blanco-Heredia et al. "Identification of Immunogenic Cytotoxic T Lymphocyte Epitopes Containing Drug Resistance Mutations in Antiretroviral Treatment-Naïve HIV-Infected Individuals". In: *PloS One* 11.1 (2016), e0147571. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0147571. pmid: 26808823.

[2] Kevin G. Buell et al. "Lifelong Antiretroviral Therapy or HIV Cure: The Benefits for the Individual Patient". In: *AIDS Care* 28.2 (Feb. 1, 2016), pp. 242–246. ISSN: 0954-0121. DOI: 10.1080/09540121.2015.1074653. pmid: 26357912. URL: https://doi.org/10.1080/09540121.2015.1074653 (visited on 04/19/2024).

[3] Dana S. Clutter et al. "HIV-1 Drug Resistance and Resistance Testing". In: *Infection, Genetics and Evolution: Journal of Molecular Epidemiology and Evolutionary Genetics in Infectious Diseases* 46 (Dec. 2016), pp. 292–307. ISSN: 1567-7257. DOI: 10.1016/j.meegid.2016.08.031. pmid: 27587334.

[4] Monique R. Ferguson et al. "HIV-1 Replication Cycle". In: *Clinics in Laboratory Medicine* 22.3 (Sept. 2002), pp. 611–635. ISSN: 0272-2712. DOI: 10.1016/s0272-2712(02)00015-x. pmid: 12244589.

[5] Victor Garcia and Roland Robert Regoes. "The Effect of Interference on the CD8+ T Cell Escape Rates in HIV". In: *Frontiers in Immunology* 5 (2015). ISSN: 1664-3224. URL: https://www.frontiersin.org/articles/10.3389/fimmu.2014.00661 (visited on 10/18/2023).

[6] Kevin M. Gibas et al. "Two-Drug Regimens for HIV Treatment". In: *The lancet. HIV* 9.12 (Dec. 2022), e868–e883. ISSN: 2352-3018. DOI: 10.1016/S2352-3018(22)00249-1. pmid: 36309038.

[7] "HIV-1 Genetic Variation and Drug Resistance Development". In: *Expert Review of Anti-infective Therapy* 11.11 (2013), pp. 1159–1178. ISSN: 1478-7210. DOI: 10.1586/14787210.2013.844649. URL: https://journals.scholarsportal.info/details/14787210/v11i0011/1159_hgvadrd.xml (visited on 10/18/2023).

[8] J. D. Hunter. "Matplotlib: A 2D Graphics Environment". In: *Computing in Science & Engineering* 9.3 (2007), pp. 90–95. DOI: 10.1109/MCSE.2007.55.

[9] Vanessa Jurtz et al. "NetMHCpan 4.0: Improved Peptide-MHC Class I Interaction Predictions Integrating Eluted Ligand and Peptide Binding Affinity Data". In: *Journal of immunology (Baltimore, Md. : 1950)* 199.9 (Nov. 1, 2017), pp. 3360–3368. ISSN: 0022-1767. DOI: 10.4049/jimmunol.1700893. pmid: 28978689. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5679736/ (visited on 04/15/2024).

[10] James G Kahn et al. "Cost-Effectiveness of Antiretroviral Therapy for Prevention". In: *Current HIV Research* 9.6 (Sept. 2011), pp. 405–415. ISSN: 1570-162X. DOI: 10.2174/157016211798038542. pmid: 21999776. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3529401/ (visited on 06/19/2023).

[11] Brendan A. Larder. "Interactions Between Drug Resistance Mutations in Human Immunodeficiency Virus Type 1 Reverse Transcriptase". In: *Journal of General Virology* 75.5 (1994), pp. 951–957. ISSN: 1465-2099. DOI: 10.1099/0022-1317-75-5-951. URL: https://www.microbiologyresearch.org/content/journal/jgv/10.1099/0022-1317-75-5-951 (visited on 10/18/2023).

[12] A. J. Leslie et al. "HIV Evolution: CTL Escape Mutation and Reversion after Transmission". In: *Nature Medicine* 10.3 (3 Mar. 2004), pp. 282–289. ISSN: 1546-170X. DOI: 10.1038/nm992. URL: https://www.nature.com/articles/nm992 (visited on 10/18/2023).

[13] Francesca Lombardi et al. "Prevalence and Factors Associated with HIV-1 Multi-Drug Resistance over the Past Two Decades in the Italian ARCA Database". In: *International Journal of Antimicrobial Agents* 57.2 (Feb. 2021), p. 106252. ISSN: 1872-7913. DOI: 10.1016/j.ijantimicag.2020.106252. pmid: 33259914.

[14] Jennifer L Mamrosh et al. "HIV Molecular Immunology 2022". In: (2022).

[15] Rosemarie D. Mason et al. "Antiretroviral Drug Resistance Mutations Sustain or Enhance CTL Recognition of Common HIV-1 Pol Epitopes". In: *Journal of Immunology (Baltimore, Md.: 1950)* 172.11 (June 1, 2004), pp. 7212–7219. ISSN: 0022-1767. DOI: 10.4049/jimmunol.172.11.7212. pmid: 15153547.

[16] Sandra M. Mueller et al. "Influence of Major HIV-1 Protease Inhibitor Resistance Mutations on CTL Recognition". In: *Journal of Acquired Immune Deficiency Syndromes (1999)* 56.2 (Feb. 1, 2011), pp. 109–117. ISSN: 1944-7884. DOI: 10.1097/QAI.0b013e3181fe946e. pmid: 21107269.

[17] Nittaya Phanuphak and Roy M. Gulick. "HIV Treatment and Prevention 2019: Current Standards of Care". In: *Current Opinion in HIV and AIDS* 15.1 (Jan. 2020), p. 4. ISSN: 1746-630X. DOI: 10.1097/COH.0000000000000588. URL: https://journals.lww.com/co-hivandaids/Abstract/2020/01000/HIV_treatment_and_prevention_2019__current.3.aspx (visited on 06/19/2023).

[18] Mohammad Arif Rahman et al. "Raltegravir and Elvitegravir-Resistance Mutation E92Q Affects HLA-B*40:02-Restricted HIV-1-specific CTL Recognition". In: *Microbes and Infection* 16.5 (May 2014), pp. 434–438. ISSN: 1769-714X. DOI: 10.1016/j.micinf.2014.03.003. pmid: 24657622.

[19] Birkir Reynisson et al. "NetMHCpan-4.1 and NetMHCIIpan-4.0: Improved Predictions of MHC Antigen Presentation by Concurrent Motif Deconvolution and Integration of MS MHC Eluted Ligand Data". In: *Nucleic Acids Research* 48.W1 (July 2, 2020), W449–W454. ISSN: 0305-1048. DOI: 10.1093/nar/gkaa379. pmid: 32406916. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7319546/ (visited on 04/15/2024).

[20] Soo-Yon Rhee et al. "HIV-1 Transmitted Drug Resistance Surveillance: Shifting Trends in Study Design and Prevalence Estimates". In: *Journal of the International AIDS Society* 23.9 (Sept. 2020), e25611. ISSN: 1758-2652. DOI: 10.1002/jia2.25611. pmid: 32936523.

[21] Douglas D. Richman et al. "HIV Evolution and Escape". In: *Transactions of the American Clinical and Climatological Association* 115 (2004), pp. 289–303. ISSN: 0065-7778. pmid: 17060974.

[22] Erik J. Schouten et al. "Antiretroviral Drug Supply Challenges in the Era of Scaling up ART in Malawi". In: *Journal of the International AIDS Society* 14 Suppl 1 (July 6, 2011), S4. ISSN: 1758-2652. DOI: 10.1186/1758-2652-14-S1-S4. pmid: 21967844.

[23] F. Douglas Scutchfield and A. S. Benenson. "AIDS Update". In: *Postgraduate Medicine* 85.4 (Mar. 1, 1989), pp. 289–304. ISSN: 0032-5481. DOI: 10.1080/00325481.1989.11700636. URL: https://doi.org/10.1080/00325481.1989.11700636 (visited on 04/19/2024).

[24] Sorana Segal-Maurer et al. "Capsid Inhibition with Lenacapavir in Multidrug-Resistant HIV-1 Infection". In: *The New England Journal of Medicine* 386.19 (May 12, 2022), pp. 1793–1803. ISSN: 1533-4406. DOI: 10.1056/NEJMoa2115542. pmid: 35544387.

[25] Paul M. Sharp and Beatrice H. Hahn. "Origins of HIV and the AIDS Pandemic". In: *Cold Spring Harbor Perspectives in Medicine:* 1.1 (Sept. 2011), a006841. ISSN: 2157-1422. DOI: 10.1101/cshperspect.a006841. pmid: 22229120. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3234451/ (visited on 06/05/2023).

[26] The pandas development team. *Pandas-Dev/Pandas: Pandas*. Version v2.2.2. Zenodo, Apr. 2024. DOI: 10.5281/zenodo.10957263. URL: https://doi.org/10.5281/zenodo.10957263.

[27] Angel Varela-Rohena et al. "Control of HIV-1 Immune Escape by CD8 T Cells Expressing Enhanced T-cell Receptor". In: *Nature Medicine* 14.12 (Dec. 2008), pp. 1390–1395. ISSN: 1078-8956, 1546-170X. DOI: 10.1038/nm.1779. URL: https://www.nature.com/articles/nm.1779 (visited on 10/18/2023).

[28] John D. Ventura. "Human Immunodeficiency Virus 1 (HIV-1): Viral Latency, the Reservoir, and the Cure". In: *The Yale Journal of Biology and Medicine* 93.4 (Sept. 30, 2020), pp. 549–560. ISSN: 0044-0086. pmid: 33005119. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7513431/ (visited on 06/04/2023).

[29] Pauli Virtanen et al. "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python". In: *Nature Methods* 17 (2020), pp. 261–272. DOI: 10.1038/s41592-019-0686-2.

[30] Michael L. Waskom. "Seaborn: Statistical Data Visualization". In: *Journal of Open Source Software* 6.60 (2021), p. 3021. DOI: 10.21105/joss.03021. URL: https://doi.org/10.21105/joss.03021.

[31] Zenda Woodman and Carolyn Williamson. "HIV Molecular Epidemiology: Transmission and Adaptation to Human Populations". In: *Current Opinion in HIV and AIDS* 4.4 (July 2009), pp. 247–252. ISSN: 1746-630X. DOI: 10.1097/COH.0b013e32832c0672. pmid: 19532060. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3351078/ (visited on 10/18/2023).